Contents

1 Introduction

Snow is one of the major component of the cryosphere. Its global distribution on both ocean and land areas strongly depends on seasonality, latitude, and elevation. It occurs in three forms, mainly: permanent snow on ice sheets and glaciers, snow on sea ice, and seasonal snow cover. The latter is estimated to account for 1.3% (summer) to 30.6% (winter) of the Northern Hemisphere land surface (Vaughan et al., 2013), with the 1967-2026 mean snow cover extent (SCE hereafter) in winter being around $39.79 \times 10^6\,\mathrm{km}^2$ (Robinson & Estilow, 2012). Sea ice is almost completely covered by snow during the winter season; its peak concentration, therefore representative of sea ice specific SCE, occurs in March in the Northern Hemisphere ($15.03 \times 10^6\,\mathrm{km}^2$) and in September in the Southern Hemisphere ($18.61 \times 10^6\,\mathrm{km}^2$) (Fetterer et al., 2025; Warren et al., 1999). Finally, perennial snow mean extent over the 2000-2022 period is estimated at $16.47 \times 10^6\,\mathrm{km}^2$ from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS), including glaciers, ice caps, Greenland and Antarctica ice sheets (Johnston et al., 2024).

These several tens of millions of snow-covered squared kilometres play a major role in freshwater resources availability, ecosystems equilibrium and, particularly, climate regulation. In fact, among its many physico-chemical properties, snow represents one of the strongest reflective materials on Earth's surface. The fraction of solar radiation delivered to a body that is diffusely reflected by it is called *albedo*; for snow, this amount is roughly close to 1, since its highest reflectivity occurs in the range of visible light, *i.e.* where solar radiation intensity peaks (A. A. Kokhanovsky & Zege, 2004). Thanks to its high albedo, snow represents of the main contributors to Earth's energy budget regulators (National Snow and Ice Data Center, 2024). Of the $340\,\mathrm{W\,m^{-2}}$ of net solar radiation incident at the top of the atmosphere, Northern Hemisphere land snow cover alone reflects approximately $2\,\mathrm{W\,m^{-2}}$ back to space, averaged over the 1979–2008 period; in May, when SCE and soil insulation are highest, this quantity can reach $4.5\,\mathrm{W\,m^{-2}}$ (Flanner et al., 2011). Without the year-average radiative forcing from land snow cover alone, Earth's surface temperature would be approximately 0.54 K higher than the current planetary mean. This estimate follows from linearising the Stefan-Boltzmann law with respect to temperature. This assumes Earth radiates as a blackbody with no feedback effects, making it a lower-bound estimate; in fact, real climate feedbacks (water vapour, lapse rate, clouds) reduce the effective restoring force, amplifying the true temperature response (**cronin**).

As the whole cryosphere in general, snow distribution on Earth is undergoing dramatic changes because of the current anthropogenic perturbation of the climate system and warming of the atmosphere. At the global scale, SCE is experiencing a negative trend since the 1980s, with a statistically significant reduction in the mean area of 5% after 1988, compared to the 1967-1988 period, in the Northern Hemisphere only, and an earlier melt onset in the spring season (Dye, 2002; Robinson & Frei, 2000). Less snow cover in both space and time, driven by rising temperatures, reduces terrestrial albedo and increases the absorption of incoming solar radiation at the surface, which in turn amplifies warming (Cubasch et al., 2001; Holland & Bitz, 2003). This mechanism is known as the positive snow-albedo feedback: it is estimated to be responsible of a reduction in Northern Hemisphere land snow cover radiative cooling forcing of $\sim 0.22\,\mathrm{W\,m^{-2}}$ from 1979 to 2007, which nearly equals the contribution of Arctic sea ice (Flanner et al., 2011). The magnitude of such feedback is further amplified when light-absorbing particles, such as dust, black carbon and microbial growth, deposit on the surface of snowpack (Skiles et al., 2018).

It is clear, at this point, that developing a proper understanding of how snow interacts with solar radiation is crucial for a better understanding of the evolution of the climate system. Snow optical properties, particularly albedo, are directly related to density and grain type, and are strongly wavelength-dependent (Bohren & Barkstrom, 1974; A. A. Kokhanovsky & Zege, 2004). Quantifying these properties involves radiative transfer calculations extended to the uppermost part of the snowpack, which must be assumed to be composed of individual grains of a given size and shape. Many studies have adopted equivalent spheres, assuming they can properly represent any other shape at a given optical diameter (Bohren & Barkstrom, 1974; Warren et al., 1999). However, recent studies demonstrated that this approach is not satisfactory when it comes to specific properties such as the single-scattering albedo – the fraction of intercepted radiation that is scattered rather than absorbed – and the bidirectional reflectance distribution function (BRDF) – the reflectance of a surface for a specific combination of illumination and viewing directions (Dumont et al., 2010; A. A. Kokhanovsky & Zege, 2004; Libois et al., 2013). For this reason, when addressing grain size, a more general, shape-independent quantity is preferable to the optical diameter: the *specific surface area* (SSA), defined as the total surface area of snow grains per unit mass – expressed in $\mathrm{m^2\,kg^{-1}}$ (International Union of Pure and Applied Chemistry (IUPAC), 2025). This quantity, used hereafter to address snow grain size, largely influences snow reflectance for wavelengths in the Near-Infrared (NIR) range, between 750 nm and 1400 nm particularly (Warren & Wiscombe, 1980). For example, a factor-6 decrease in the SSA (*e.g.*, from 60 to $10\,\mathrm{m^2\,kg^{-1}}$) can approximately reduce albedo of 15% at 1000 nm. In general, albedo decreases monotonically with SSA and can be modelled as an exponential function of $\mathrm{SSA}^{-1/2}$ (A. A. Kokhanovsky & Zege, 2004).

Snow grain size dynamics are therefore a key indicator of how snowpack optical properties evolve over time and space, making their tracking essential. Changes in grain size are primarily driven by processes such as metamorphism, wind drift, and precipitation.

When transformations in size and shape occur below the melting point ($T = 0\,°\mathrm{C}$), the metamorphism is dry: the grain size grows at a rate defined by the respective increase in temperature and temperature gradient (Colbeck, 1982). Instead, if snow is at the melting point, the liquid water content directly drives the size increase, which is faster than in the the dry case – we talk about wet metamorphism, in this case (Wakahama, 1968). Anyway, independently of the type of metamorphism, the change in size is always positive and irreversible (**find a cit**).

Transportation by the wind occurs when its velocity breaks snow cohesion and lifts grains. This is more likely for fresh-fallen snow than for old, metamorphosed snow (Lenaerts et al., 2017). This process, called wind

drift, varies with wind speed: saltation occurs under weak winds, with snow skipping a few centimeters above the surface, while stronger winds create snow plumes, lifting grains up to hundreds of meters – the so-called blowing snow (Adok, 1977; World Meteorological Organization, 2024). Wind drift has two opposing effects on grain size. Erosion of the snowpack's upper layers exposes older, coarser, and less reflective layers (Lenaerts et al., 2017). Conversely, wind fractures grains through collisions, so newly formed deposits typically consist of finer grains (Domine et al., 2009).
Snow fall is generally responsible of depositing fresh and more fine-grained snow at surface, potentially covering already metamorphosed and coarser layers, thus increasing SSA an albedo (Domine et al., 2009).

This study is focused on the temporal evolution of snow and its albedo over the whole Antarctica, where all the processes act and interplay in a complex manner, over a wide range of climatic and geographical contexts. The Antarctic Ice Sheet (AIS), expanding for 14×10^6 km^2 over 98% of the continent, is almost completely snow-covered year-round. As a polar region, it plays a massive role in positive climate feedback. At the same time, compared to the Arctic, Antarctica presents unique challenges for the exploration of surface radiative properties.
Being such a harsh and inaccessible place, it has only allowed *in situ* investigations limited to regional scales and relatively short time coverage. This has led the scientific community to rely on satellite data, which inherits higher uncertainties and reduced spatio-temporal resolution, from missions such as AVHRR, MODIS, and CLARA-A2.1-SAL (**sunChangesAntarcticsSummer2023**; Calleja et al., 2019).
With this work, we aim to improve the understanding of surface albedo evolution in Antarctica by combining *in situ* measurements with satellite observations. Using data from automated albedometers installed at several locations on the AIS, we will produce time series of locally measured specific surface area (SSA) and compare them with satellite-retrieved grain size. Measurement stations were deployed in two primary locations: on the coast near the French station Dumont d'Urville, and in a megadune field, with one albedometer on the erosion area and another on the accumulation area of a dune.

The overarching goals of this analysis are:

- to validate a satellite product, ESA's Sentinel-3 Ocean and Land Color Instrument, enabling long-term and continent-wide studies of snow albedo at a relatively fine resolution (~ 300 m), compared to other missions (e.g., MODIS);
- to deepen the understanding of processes driving temporal albedo changes at the local scale, particularly on megadunes, using an unprecedented 6-year continuous *in situ* record of snow albedo.

The analysis is structured as follows:

1. **Processing of *in situ* data**: upgrading and optimizing the existing workflow developed by Arioli et al., 2023; this includes both processing data from the main instrument (Multiband and relative weather stations) and from Solalb, a second albedometer used as as external reference to asses the quality of Multiband and Sentinel-3 performaces;

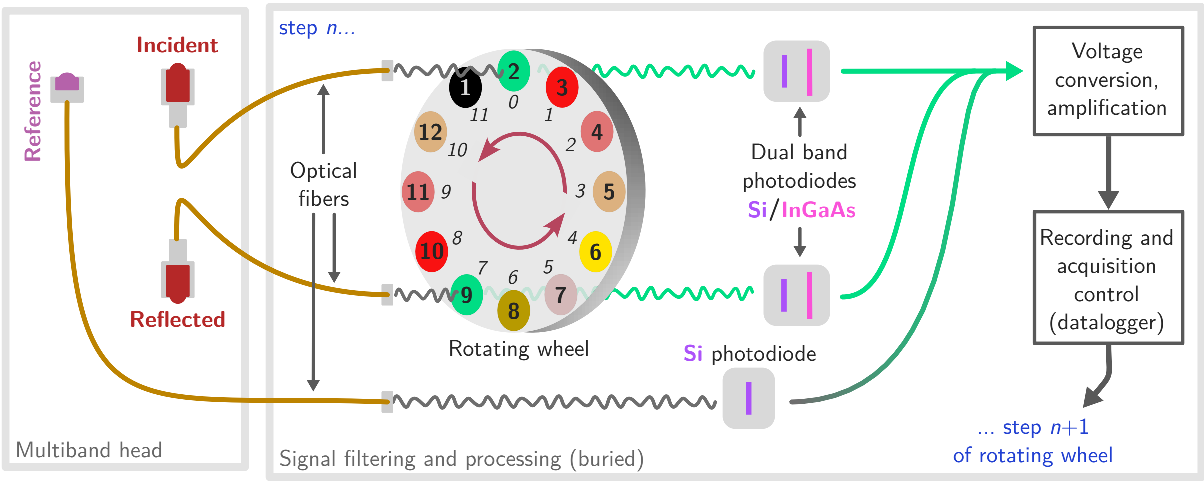

Figure 1: Structure of Multiband | Long caption

2. **Processing of Sentinel-3 data**: downloading the frames and executing the SICE 1.6 algorithm, as described by A. Kokhanovsky et al., 2019;
3. **Comparison of the time series**: performing corrections to address possible gaps between the two datasets and interpreting the retrieved albedo dynamics geoscientifically.

II Material and methods

1 Theoretical backgrounds

!!! This risks to become "textbook"-like - then, must be SUPER ADAPTED TO MY SPECIFIC GOAL AND BE VERY PERSONAL AND NOT GENERAL AT ALL - PASSIONATING ! write it after matmethods and results

2 Study area: an overview of the station

3 Material and instruments

3.1 *In situ*: the Multiband albedometer

The main instrument used in this study is the Multiband albedometer, developed at Grenoble's Institute for Geosciences of the Environment (IGE) and fully described in Arioli et al., 2023. It is a robust automated optical radiometer capable of acquiring solar radiation across a discrete set of bands, from the visible to the short-wave infrared, which are essential for retrieving snow surface albedo. Similar to previously developed albedometers (e.g., Autosolexs, Solalb; see Picard et al., 2016), the Multiband consists of a set of metal gantries. One perfectly leveled pole projects the instrument head horizontally at ~ 3 m from the installation point and at a height of ~ 2 m above the snow surface, typically oriented southward to minimize shadowing from other components. The head features two custom cosine-shaped light collectors, one pointing upward and the other downward, to measure incident and reflected irradiance, respectively.

These collectors, meticulously designed by Picard et al., 2016, incorporate a bumped Teflon surface to maximize performance at any solar zenith angle (SZA), including grazing angles, which are critical for accurate radiation measurements.

The signal is transmitted to the instrument's recording unit via two Ocean Optics QP800-VIS-NIR optical fibers, each 6 m long with a 800 µm core diameter. Unlike earlier albedometers, where optical fibers directly sent the signal to a spectrometer for high-resolution (3 nm) spectral recording, the Multiband employs a more energy-efficient approach. The broadband radiation transmitted by the fibers passes through a set of polarizing filters mounted on a $\sim$ 20 cm wheel with 12 slots. These filters are Edmund Optics' TECHSPEC® hard-coated optical density (OD) 4.0, 25 nm bandpass filters, specific for wavelengths of 500, 600, 800, 925, 1050, 1300, and 1550 nm. During a full measurement cycle (30 s), the wheel rotates to acquire spectral measurements for each selected wavelength from both the incident and reflected signals. These signals are transmitted to dual-band Silicon (Si) / Indium-Arsenide-Gallium (InGaAs) photodiodes (DSD2, Thorlabs). Measurements on the two channels are simultaneous, and the wheel includes duplicate filters to ensure synchronous recordings at the most relevant wavelengths, as well as one capped slot to measure the dark current (*i.e.* background thermal noise) – see Figure 1 for details. The signals registered by the dual photodiodes are amplified by a high-gain electronic board. Finally, a Campbell Scientific CR1000x data logger records the spectral radiation. Since the duration of a full measurement cycle is long enough for atmospheric conditions to potentially change, a measure of ambient light is also recorded at every step: a third upward-facing optical fiber transmits incoming radiation to a Si photodiode (FDS100, Thorlabs). This photodiode acquires a wideband measurement in the 350–1100 nm range, referred to as the reference. Its signal undergoes the same voltage amplification and recording steps as the spectral measurements.

The entire system's electronics are powered by a solar panel. The Multiband has demonstrated its durability by acquiring continuous measurements without interruption at each of the stations where it was installed.

3.2 *In situ*: the Solalb albedometer

3.3 *In situ*: automatic weather stations (AWS)

3.4 Satellite imagery: Sentinel-3 Ocean Land and Colour Instrument

3.4.a Generalities

Orbit, resolution, swath, revisit time etc.

3.4.b Data availability

4 Methods

4.1 Processing of Multiband data

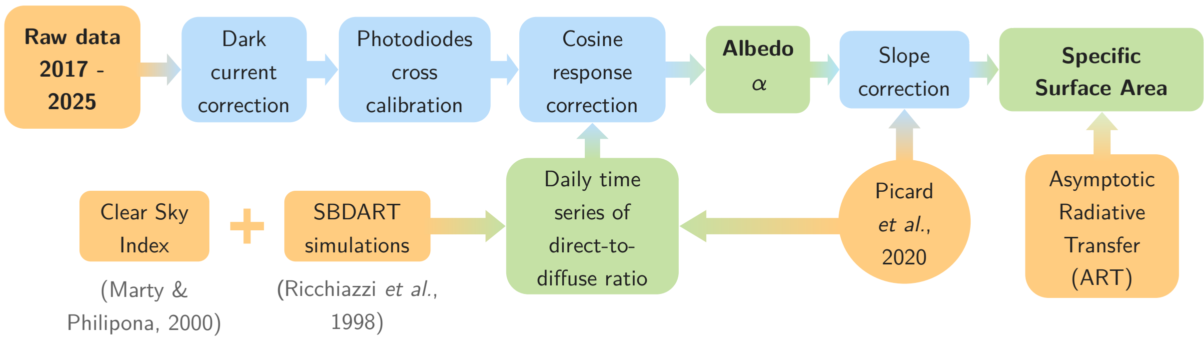

Figure 2: Workflow of Multiband processing | Modified after Arioli et al., 2023

4.1.a Overview of the existing algorithm

1. **Raw data filtering**

a

2. **Dark current**

   b

3. **Cross-calibration**

   c

4. **Cosine response correction**

   d

5. **Slope correction**

   e

4.1.b New implementations

Extended the processing to newly installed stations in megadune field...

Direct-to-diffuse light ratio

Clear Sky Index (CSI)

Diagnostic techniques

- **Calibration files**

- **Inclinometer time series**

- **Sub-daily variability in the signal**

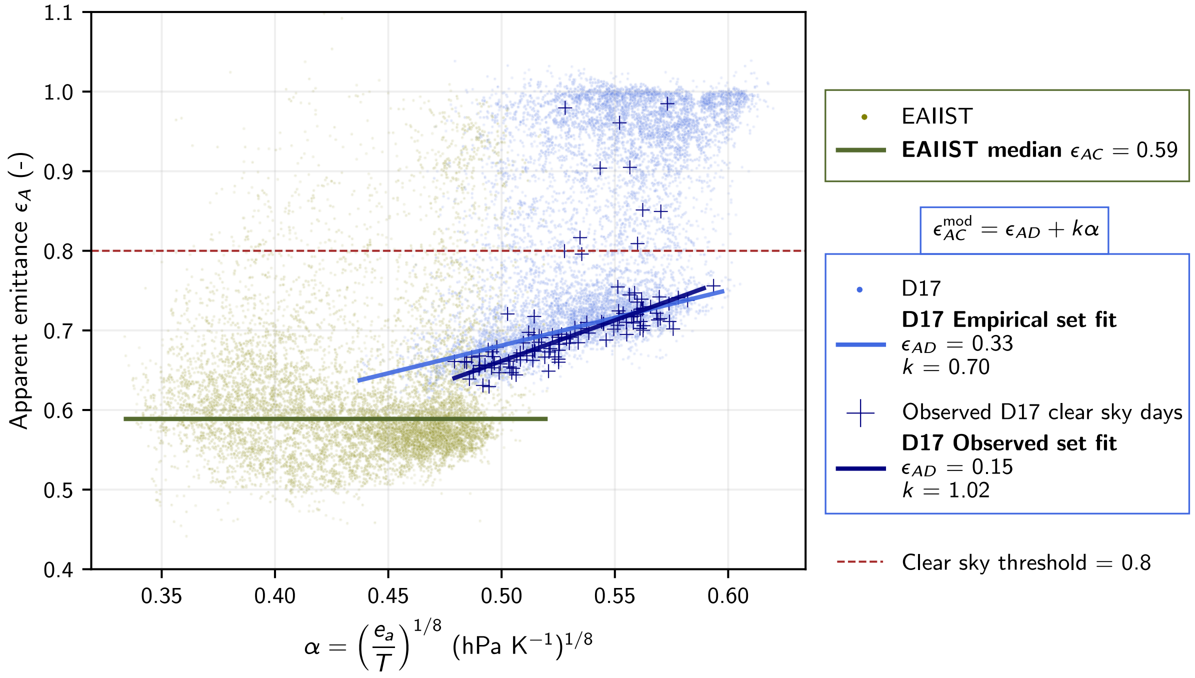

Figure 3: Apparent emissivity | Test figure

4.2 Processing of Solalb data

4.3 Processing of S-3 OLCI data

4.3.a Implementation of SICE 1.6 utilities

4.3.b Diagnostic techniques

III Results

5 Atmospheric conditions: emissivity analyisis

CSI results ? Dir diff ?

6 Time series from Multiband and S-3 OLCI

DDU very cool, EAIIST not so very cool, then...

7 Diagnostic of the results for EAIIST stations

... then here we go

7.1 Multiband issues

7.2 OLCI issues

7.2.a OLCI and Solalb: a comparison

Do not simply list the results, it can be boring If the comclusion from a result is very fast, just put it in the results

IV Discussion

8 Considerations on the performance of the instruments

9 Geoscientific interpretation of the time series

Despite the imperfection of the results and the technical issues, some interesting interpretations about the signal recorded from both the *in situ* equipment and the satellite can be drawn.

V Conclusion

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